

Supplementary Material for “Two-component treatment comparisons for continuous outcomes truncated by a substantive event”

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A Proofs for main text results

Proposition 1 (Parametric likelihood ratio calibration). *Under Assumption 2, if $(\mu_{\delta 0}, \beta_{\delta 0})$ is the true component contrast, then*

$$W_n^{\text{par}}(\mu_{\delta 0}, \beta_{\delta 0}) \xrightarrow{d} \chi_2^2.$$

In particular, under the joint null, $W_n^{\text{par}}(0, 0) \xrightarrow{d} \chi_2^2$.

Proof of Proposition 1. The working likelihood factorizes into a Gaussian mean component and a Bernoulli component with variation-independent parameter blocks. Profiling at a fixed component contrast therefore separates exactly into the two displayed component deviances.

For the Bernoulli component, ordinary profile likelihood theory gives a squared standardized efficient score plus $o_p(1)$. For the Gaussian component, profiling σ^2 gives the residual sum of squares deviance. The correct conditional mean, common conditional variance, information convergence, and Lindeberg condition imply that its expansion at the true mean contrast is also a squared standardized efficient score plus $o_p(1)$. Write

$$x_{\beta i} = (1, R_i, h_{\beta}(X_i)^{\top})^{\top}, \quad x_{\mu i} = (1, R_i, h_{\mu}(X_i)^{\top})^{\top}.$$

The Bernoulli score contribution is $S_{\beta i} = x_{\beta i}(A_i - \pi_i)$. Define the Gaussian mean score contribution by

$$S_{\mu i} = \begin{cases} \sigma_0^{-2} x_{\mu i}(Y_i - \mu_{R_i}(X_i)), & A_i = 1, \\ 0, & A_i = 0. \end{cases}$$

Its conditional covariance with $S_{\beta i}$ is zero because

$$\mathbb{E}[S_{\beta i} S_{\mu i}^{\top} \mid R_i, X_i] = \sigma_0^{-2} \pi_i (1 - \pi_i) x_{\beta i} x_{\mu i}^{\top} \mathbb{E}[Y_i - \mu_{R_i}(X_i) \mid A_i = 1, R_i, X_i] = 0.$$

Linear nuisance projection preserves this zero covariance. The joint central limit theorem therefore gives two independent standard Normal limits, and the sum of their squares converges to χ_2^2 . The same argument applies at any true interior component contrast, including $(0, 0)$ under the null. $\square$

Proposition 2 (Covariate-adjusted semiparametric likelihood ratio test). *Under Assumption 3, if $(\mu_{\delta 0}, \beta_{\delta 0})$ is the true component contrast, then*

$$\widetilde{W}_n^{\text{semi}}(\mu_{\delta 0}, \beta_{\delta 0}) = \widetilde{W}_{Y,n}^{\text{prof}}(\mu_{\delta 0}) + W_{A,n}^{\text{prof}}(\beta_{\delta 0}) \xrightarrow{d} \chi_2^2.$$

In particular, under the null hypothesis $\mu_{\delta} = \beta_{\delta} = 0$, $\widetilde{W}_n^{\text{semi}}(0, 0) \xrightarrow{d} \chi_2^2$.

Proof of Proposition 2. By the profiled Bernoulli likelihood regularity in Assumption 3, the expansion at the true contrast gives

$$W_{A,n}^{\text{prof}}(\beta_{\delta 0}) = U_{A,n}^2 + o_p(1), \quad U_{A,n} \xrightarrow{d} N(0, 1).$$

Similarly, the profiled estimating equation empirical likelihood regularity conditions give the empirical likelihood expansion at the true mean contrast,

$$\widetilde{W}_{Y,n}^{\text{prof}}(\mu_{\delta 0}) = U_{Y,n}^2 + o_p(1), \quad U_{Y,n} \xrightarrow{d} N(0, 1).$$

The joint central limit theorem and the zero cross covariance established after Assumption 3 imply

$$(U_{A,n}, U_{Y,n})^\top \xrightarrow{d} N_2(0, I_2).$$

Therefore,

$$\widetilde{W}_n^{\text{semi}}(\mu_{\delta 0}, \beta_{\delta 0}) = W_{A,n}^{\text{prof}}(\beta_{\delta 0}) + \widetilde{W}_{Y,n}^{\text{prof}}(\mu_{\delta 0}) = U_{A,n}^2 + U_{Y,n}^2 + o_p(1) \xrightarrow{d} \chi_2^2.$$

The null statement is the special case $(\mu_{\delta 0}, \beta_{\delta 0}) = (0, 0)$. $\square$

Proposition 3 (Projected likelihood ratio range for the coded mean contrast). *Under Assumption 4, the projected interval in Equation (5) has asymptotic coverage at least $1 - \alpha$ for $d_0 = D_{e_0}(\zeta_0)$.*

$$\liminf_{n \rightarrow \infty} P(d_0 \in \text{CI}_{1-\alpha}^{\Delta, \text{proj}}) \geq 1 - \alpha.$$

Proof of Proposition 3. Assumption 4 gives

$$\mathbb{W}_n(\zeta_0) = \widehat{h}_n^\top I_\zeta \widehat{h}_n + o_p(1) \xrightarrow{d} \chi_4^2,$$

so $\mathcal{C}_{4,n}(1 - \alpha)$ contains ζ_0 with limiting probability $1 - \alpha$. Whenever it contains ζ_0 , its image contains $D_{e_0}(\zeta_0) = d_0$. The claimed lower bound follows. $\square$

Proposition 4 (Profile likelihood ratio interval for the coded mean contrast). *Under Assumption 4, $T_n^D(d_0) \xrightarrow{d} \chi_1^2$, and the profile interval in Equation (7) has asymptotic coverage $1 - \alpha$ for d_0 .*

Proof of Proposition 4. Let $a = \nabla D_{e_0}(\zeta_0) \neq 0$ and write $h = \sqrt{n}(\zeta - \zeta_0)$. The localization part of Assumption 4 applies to an approximate minimizer of $T_n^D(d_0)$ because $T_n^D(d_0) \leq \mathbb{W}_n(\zeta_0) = O_p(1)$. Uniform differentiability on bounded local sets gives

$$D_{e_0}(\zeta_0 + h/\sqrt{n}) = d_0 + n^{-1/2} a^\top h + o(n^{-1/2}).$$

Let the exact local constraint set be the set of bounded h for which the left side equals d_0 , and let its limiting tangent set be $\{h : a^\top h = 0\}$. Every bounded sequence in the exact set is within $o(1)$ of the tangent set. Conversely, since $\partial D_{e_0}/\partial \mu_\delta = \pi_1 > 0$, the implicit function theorem gives, for every bounded point in the tangent set, a point in the exact set that differs from it by $o(1)$. The two constraint sets therefore give the same limiting quadratic infimum. Consequently,

$$T_n^D(d_0) = \inf_{a^\top h=0} (h - \widehat{h}_n)^\top I_\zeta (h - \widehat{h}_n) + o_p(1) = \frac{(a^\top \widehat{h}_n)^2}{a^\top I_\zeta^{-1} a} + o_p(1).$$

The final expression converges in distribution to χ_1^2 . Inversion at the continuous χ_1^2 cutoff gives the coverage result, and the exact image identity above gives the reported endpoint representation. $\square$

B Supplementary simulation results

This appendix gives additional design details for the simulation study in Section 3 and reports the full numerical rejection rates underlying the main text figures.

For each cell defined by scenario, effect level, and sample size, the random-number seed was set to 20260416 plus the cell index minus one. Cells were ordered by scenario, then effect level, then sample size. Sample size varied fastest within each combination of scenario and effect level. Each cell used 25,000 Monte Carlo repetitions. Each replicate was analyzed using the Wilcoxon rank-sum test, Welch’s two-sample t -test, the parametric likelihood ratio test, and the semiparametric likelihood ratio test. Rejection was recorded when the method returned a finite p -value below 0.05. A method failure was recorded when the method did not return a finite p -value.

Figure 1 gives the full null calibration display for the $h = 0$ cells. The figure and tables report rejection percentages for 5% tests. In the scenario tables, $h = 0$ rows assess the common null configuration and $h > 0$ rows report rejection percentages under the displayed data-generating configurations. Because the procedures generally target different null hypotheses, the latter are not power comparisons for a common estimand. LRT and SPLRT denote the parametric and semiparametric likelihood ratio tests. The column “Expected $\Delta^{(0)}$ ” is the expected coded mean contrast for atom code $e_0 = 0$.

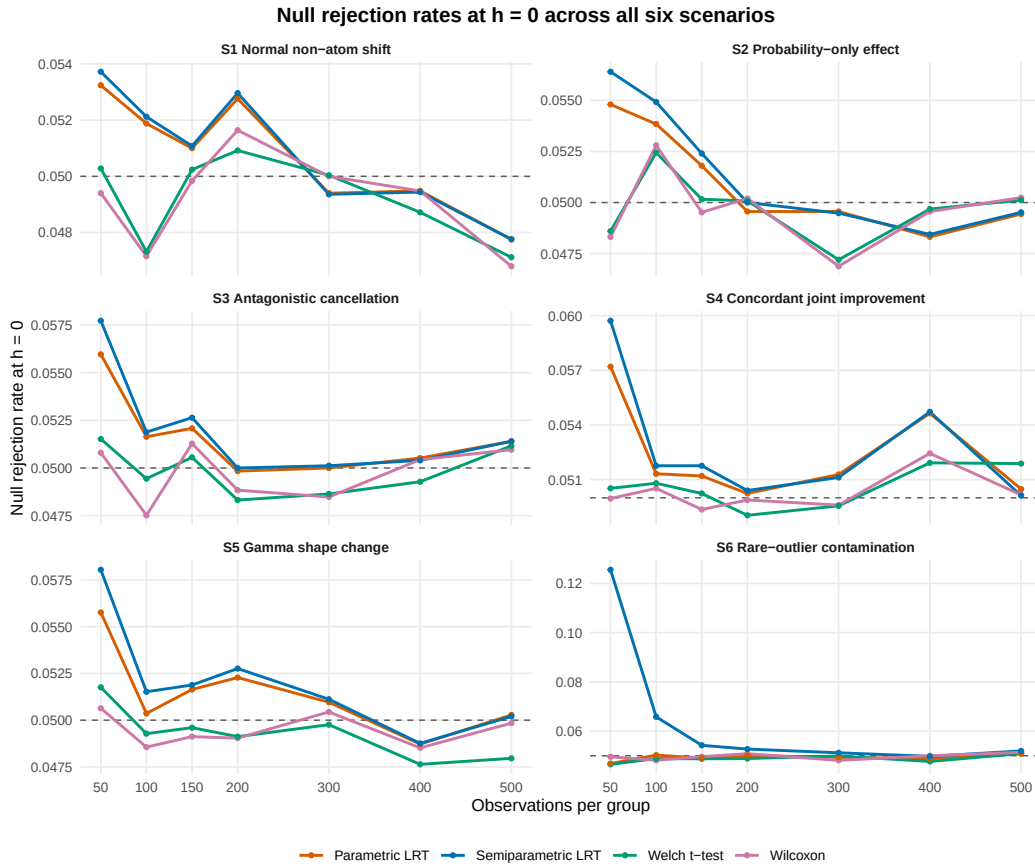


Figure 1: Null rejection rates at effect level $h = 0$ across all six simulation scenarios. The dashed line marks 0.05.

Table 1: Scenario S1 (Normal non-atom shift, low atom probability): estimated rejection percentages by sample size and effect level

n_{arm}	h	Effect	Expected $\Delta^{(0)}$	Wilcoxon (%)	Welch t -test (%)	LRT (%)	SPLRT (%)
50	0	$\delta_h = 0.00$	0.000	4.94	5.03	5.32	5.37
100	0	$\delta_h = 0.00$	0.000	4.72	4.73	5.19	5.21
150	0	$\delta_h = 0.00$	0.000	4.98	5.02	5.10	5.11
200	0	$\delta_h = 0.00$	0.000	5.16	5.09	5.28	5.30
300	0	$\delta_h = 0.00$	0.000	5.00	5.00	4.94	4.94
400	0	$\delta_h = 0.00$	0.000	4.95	4.87	4.95	4.94
500	0	$\delta_h = 0.00$	0.000	4.68	4.71	4.78	4.78
50	1	$\delta_h = 0.20$	0.170	10.65	9.25	12.44	12.56
100	1	$\delta_h = 0.20$	0.170	16.89	13.26	20.16	20.17
150	1	$\delta_h = 0.20$	0.170	23.37	17.78	27.97	27.97
200	1	$\delta_h = 0.20$	0.170	29.32	21.86	35.98	36.01
300	1	$\delta_h = 0.20$	0.170	40.84	30.20	50.76	50.78
400	1	$\delta_h = 0.20$	0.170	52.36	39.04	64.54	64.60
500	1	$\delta_h = 0.20$	0.170	60.72	45.90	74.45	74.40
50	2	$\delta_h = 0.35$	0.298	22.98	17.55	28.78	29.02
100	2	$\delta_h = 0.35$	0.298	41.03	30.20	51.88	52.00
150	2	$\delta_h = 0.35$	0.298	56.84	41.98	70.91	70.96
200	2	$\delta_h = 0.35$	0.298	69.77	53.36	83.26	83.22
300	2	$\delta_h = 0.35$	0.298	85.27	69.76	95.02	95.03
400	2	$\delta_h = 0.35$	0.298	93.56	81.58	98.80	98.80
500	2	$\delta_h = 0.35$	0.298	97.37	89.66	99.76	99.76
50	3	$\delta_h = 0.50$	0.425	41.04	29.95	52.75	53.05
100	3	$\delta_h = 0.50$	0.425	69.14	51.71	83.67	83.70
150	3	$\delta_h = 0.50$	0.425	85.45	69.44	95.45	95.44
200	3	$\delta_h = 0.50$	0.425	93.76	81.30	98.95	98.95
300	3	$\delta_h = 0.50$	0.425	98.86	93.72	99.94	99.94
400	3	$\delta_h = 0.50$	0.425	99.88	97.97	100.00	100.00
500	3	$\delta_h = 0.50$	0.425	99.98	99.40	100.00	100.00

Table 2: Scenario S2 (effect only on probability outside atom): estimated rejection percentages by sample size and effect level

n_{arm}	h	Effect	Expected $\Delta^{(0)}$	Wilcoxon (%)	Welch t -test (%)	LRT (%)	SPLRT (%)
50	0	$\pi_{1,h} = 0.45$	0.00	4.83	4.86	5.48	5.64
100	0	$\pi_{1,h} = 0.45$	0.00	5.28	5.24	5.38	5.49
150	0	$\pi_{1,h} = 0.45$	0.00	4.95	5.02	5.18	5.24
200	0	$\pi_{1,h} = 0.45$	0.00	5.02	5.01	4.96	5.00
300	0	$\pi_{1,h} = 0.45$	0.00	4.69	4.72	4.96	4.95
400	0	$\pi_{1,h} = 0.45$	0.00	4.96	4.97	4.83	4.84
500	0	$\pi_{1,h} = 0.45$	0.00	5.02	5.01	4.94	4.95
50	1	$\pi_{1,h} = 0.50$	0.15	7.70	7.74	7.74	7.91
100	1	$\pi_{1,h} = 0.50$	0.15	9.90	9.76	9.04	9.04
150	1	$\pi_{1,h} = 0.50$	0.15	13.13	12.89	11.96	12.01
200	1	$\pi_{1,h} = 0.50$	0.15	15.28	14.79	13.08	13.09
300	1	$\pi_{1,h} = 0.50$	0.15	20.23	19.51	17.86	17.88
400	1	$\pi_{1,h} = 0.50$	0.15	26.26	25.10	22.88	22.90
500	1	$\pi_{1,h} = 0.50$	0.15	31.61	30.40	27.64	27.64
50	2	$\pi_{1,h} = 0.55$	0.30	14.73	14.51	13.93	14.15
100	2	$\pi_{1,h} = 0.55$	0.30	25.27	24.69	23.11	23.11
150	2	$\pi_{1,h} = 0.55$	0.30	35.54	34.42	31.94	31.98
200	2	$\pi_{1,h} = 0.55$	0.30	45.40	43.97	41.52	41.54
300	2	$\pi_{1,h} = 0.55$	0.30	61.90	60.21	58.91	58.90
400	2	$\pi_{1,h} = 0.55$	0.30	74.22	72.58	71.88	71.86
500	2	$\pi_{1,h} = 0.55$	0.30	83.02	81.57	81.38	81.39
50	3	$\pi_{1,h} = 0.60$	0.45	27.30	27.02	25.59	25.84
100	3	$\pi_{1,h} = 0.60$	0.45	49.18	48.44	46.92	47.00
150	3	$\pi_{1,h} = 0.60$	0.45	66.08	64.98	64.40	64.39
200	3	$\pi_{1,h} = 0.60$	0.45	78.67	77.90	77.57	77.59
300	3	$\pi_{1,h} = 0.60$	0.45	92.17	91.39	92.10	92.10
400	3	$\pi_{1,h} = 0.60$	0.45	97.29	97.01	97.50	97.50
500	3	$\pi_{1,h} = 0.60$	0.45	99.09	99.00	99.30	99.30

Table 3: Scenario S3 (antagonistic cancellation): estimated rejection percentages by sample size and effect level

n_{arm}	h	Effect	Expected $\Delta^{(0)}$	Wilcoxon (%)	Welch t -test (%)	LRT (%)	SPLRT (%)
50	0	$\pi_{1,h} = 0.500, 1.5/\pi_{1,h} = 3.000$	0	5.08	5.15	5.60	5.77
100	0	$\pi_{1,h} = 0.500, 1.5/\pi_{1,h} = 3.000$	0	4.75	4.94	5.16	5.19
150	0	$\pi_{1,h} = 0.500, 1.5/\pi_{1,h} = 3.000$	0	5.13	5.06	5.21	5.26
200	0	$\pi_{1,h} = 0.500, 1.5/\pi_{1,h} = 3.000$	0	4.88	4.83	4.98	5.00
300	0	$\pi_{1,h} = 0.500, 1.5/\pi_{1,h} = 3.000$	0	4.85	4.86	5.00	5.01
400	0	$\pi_{1,h} = 0.500, 1.5/\pi_{1,h} = 3.000$	0	5.04	4.93	5.05	5.04
500	0	$\pi_{1,h} = 0.500, 1.5/\pi_{1,h} = 3.000$	0	5.10	5.12	5.14	5.14
50	1	$\pi_{1,h} = 0.525, 1.5/\pi_{1,h} = 2.857$	0	5.10	5.16	8.12	8.39
100	1	$\pi_{1,h} = 0.525, 1.5/\pi_{1,h} = 2.857$	0	4.92	4.94	10.43	10.46
150	1	$\pi_{1,h} = 0.525, 1.5/\pi_{1,h} = 2.857$	0	4.81	4.80	13.06	13.12
200	1	$\pi_{1,h} = 0.525, 1.5/\pi_{1,h} = 2.857$	0	5.01	4.99	16.37	16.44
300	1	$\pi_{1,h} = 0.525, 1.5/\pi_{1,h} = 2.857$	0	4.90	4.86	21.46	21.48
400	1	$\pi_{1,h} = 0.525, 1.5/\pi_{1,h} = 2.857$	0	5.15	5.03	28.53	28.52
500	1	$\pi_{1,h} = 0.525, 1.5/\pi_{1,h} = 2.857$	0	5.08	4.99	34.50	34.49
50	2	$\pi_{1,h} = 0.550, 1.5/\pi_{1,h} = 2.727$	0	4.74	4.92	15.72	16.10
100	2	$\pi_{1,h} = 0.550, 1.5/\pi_{1,h} = 2.727$	0	5.33	5.22	27.74	27.78
150	2	$\pi_{1,h} = 0.550, 1.5/\pi_{1,h} = 2.727$	0	5.19	5.06	39.20	39.23
200	2	$\pi_{1,h} = 0.550, 1.5/\pi_{1,h} = 2.727$	0	5.12	4.99	49.88	49.83
300	2	$\pi_{1,h} = 0.550, 1.5/\pi_{1,h} = 2.727$	0	5.51	5.10	68.10	68.15
400	2	$\pi_{1,h} = 0.550, 1.5/\pi_{1,h} = 2.727$	0	5.60	5.09	80.76	80.78
500	2	$\pi_{1,h} = 0.550, 1.5/\pi_{1,h} = 2.727$	0	5.56	5.10	89.15	89.10
50	3	$\pi_{1,h} = 0.600, 1.5/\pi_{1,h} = 2.500$	0	5.41	5.06	45.44	45.90
100	3	$\pi_{1,h} = 0.600, 1.5/\pi_{1,h} = 2.500$	0	5.71	5.20	76.60	76.64
150	3	$\pi_{1,h} = 0.600, 1.5/\pi_{1,h} = 2.500$	0	5.65	4.88	91.47	91.49
200	3	$\pi_{1,h} = 0.600, 1.5/\pi_{1,h} = 2.500$	0	6.06	5.08	97.12	97.12
300	3	$\pi_{1,h} = 0.600, 1.5/\pi_{1,h} = 2.500$	0	6.32	4.93	99.78	99.78
400	3	$\pi_{1,h} = 0.600, 1.5/\pi_{1,h} = 2.500$	0	6.46	5.04	99.97	99.97
500	3	$\pi_{1,h} = 0.600, 1.5/\pi_{1,h} = 2.500$	0	7.22	5.14	100.00	100.00

Table 4: Scenario S4 (concordant joint improvement): estimated rejection percentages by sample size and effect level

n_{arm}	h	Effect	Expected $\Delta^{(0)}$	Wilcoxon (%)	Welch t -test (%)	LRT (%)	SPLRT (%)
50	0	$\pi_{1,h} = 0.50, 3 + 0.15h = 3.00$	0.000	5.00	5.05	5.72	5.97
100	0	$\pi_{1,h} = 0.50, 3 + 0.15h = 3.00$	0.000	5.05	5.08	5.13	5.18
150	0	$\pi_{1,h} = 0.50, 3 + 0.15h = 3.00$	0.000	4.94	5.02	5.12	5.18
200	0	$\pi_{1,h} = 0.50, 3 + 0.15h = 3.00$	0.000	4.99	4.90	5.02	5.04
300	0	$\pi_{1,h} = 0.50, 3 + 0.15h = 3.00$	0.000	4.96	4.96	5.13	5.11
400	0	$\pi_{1,h} = 0.50, 3 + 0.15h = 3.00$	0.000	5.24	5.19	5.46	5.47
500	0	$\pi_{1,h} = 0.50, 3 + 0.15h = 3.00$	0.000	5.02	5.19	5.05	5.01
50	1	$\pi_{1,h} = 0.55, 3 + 0.15h = 3.15$	0.233	10.04	10.37	9.84	10.07
100	1	$\pi_{1,h} = 0.55, 3 + 0.15h = 3.15$	0.233	15.71	16.25	14.39	14.47
150	1	$\pi_{1,h} = 0.55, 3 + 0.15h = 3.15$	0.233	21.30	21.88	19.48	19.54
200	1	$\pi_{1,h} = 0.55, 3 + 0.15h = 3.15$	0.233	26.52	27.40	24.43	24.40
300	1	$\pi_{1,h} = 0.55, 3 + 0.15h = 3.15$	0.233	37.70	38.72	34.90	34.93
400	1	$\pi_{1,h} = 0.55, 3 + 0.15h = 3.15$	0.233	48.14	49.66	45.59	45.53
500	1	$\pi_{1,h} = 0.55, 3 + 0.15h = 3.15$	0.233	56.48	57.87	54.23	54.19
50	2	$\pi_{1,h} = 0.60, 3 + 0.15h = 3.30$	0.480	27.45	28.01	25.71	26.04
100	2	$\pi_{1,h} = 0.60, 3 + 0.15h = 3.30$	0.480	48.84	49.53	46.02	46.10
150	2	$\pi_{1,h} = 0.60, 3 + 0.15h = 3.30$	0.480	66.29	66.94	64.03	63.99
200	2	$\pi_{1,h} = 0.60, 3 + 0.15h = 3.30$	0.480	78.16	78.66	76.47	76.44
300	2	$\pi_{1,h} = 0.60, 3 + 0.15h = 3.30$	0.480	91.95	92.20	91.80	91.84
400	2	$\pi_{1,h} = 0.60, 3 + 0.15h = 3.30$	0.480	97.39	97.56	97.56	97.56
500	2	$\pi_{1,h} = 0.60, 3 + 0.15h = 3.30$	0.480	99.20	99.27	99.33	99.33
50	3	$\pi_{1,h} = 0.65, 3 + 0.15h = 3.45$	0.743	55.57	55.71	52.62	53.11
100	3	$\pi_{1,h} = 0.65, 3 + 0.15h = 3.45$	0.743	84.54	84.45	83.25	83.26
150	3	$\pi_{1,h} = 0.65, 3 + 0.15h = 3.45$	0.743	95.45	95.40	95.32	95.32
200	3	$\pi_{1,h} = 0.65, 3 + 0.15h = 3.45$	0.743	98.77	98.69	98.76	98.74
300	3	$\pi_{1,h} = 0.65, 3 + 0.15h = 3.45$	0.743	99.92	99.92	99.91	99.91
400	3	$\pi_{1,h} = 0.65, 3 + 0.15h = 3.45$	0.743	100.00	100.00	100.00	100.00
500	3	$\pi_{1,h} = 0.65, 3 + 0.15h = 3.45$	0.743	100.00	100.00	100.00	100.00

Table 5: Scenario S5 (Gamma shape change with non-atom mean shift): estimated rejection percentages by sample size and effect level

n_{arm}	h	Effect	Expected $\Delta^{(0)}$	Wilcoxon (%)	Welch t -test (%)	LRT (%)	SPLRT (%)
50	0	$k_h = 9.0, m_h = 3.0$	0.00	5.06	5.18	5.58	5.80
100	0	$k_h = 9.0, m_h = 3.0$	0.00	4.86	4.93	5.04	5.15
150	0	$k_h = 9.0, m_h = 3.0$	0.00	4.91	4.96	5.16	5.19
200	0	$k_h = 9.0, m_h = 3.0$	0.00	4.90	4.91	5.23	5.28
300	0	$k_h = 9.0, m_h = 3.0$	0.00	5.04	4.98	5.10	5.11
400	0	$k_h = 9.0, m_h = 3.0$	0.00	4.85	4.76	4.87	4.88
500	0	$k_h = 9.0, m_h = 3.0$	0.00	4.98	4.80	5.03	5.02
50	1	$k_h = 6.0, m_h = 3.2$	0.13	5.96	6.63	8.91	9.80
100	1	$k_h = 6.0, m_h = 3.2$	0.13	6.20	7.92	12.72	13.58
150	1	$k_h = 6.0, m_h = 3.2$	0.13	6.85	9.26	16.30	17.12
200	1	$k_h = 6.0, m_h = 3.2$	0.13	8.29	11.55	21.71	22.67
300	1	$k_h = 6.0, m_h = 3.2$	0.13	9.30	14.70	30.54	31.38
400	1	$k_h = 6.0, m_h = 3.2$	0.13	11.04	18.03	40.04	40.98
500	1	$k_h = 6.0, m_h = 3.2$	0.13	12.30	21.38	48.68	49.41
50	2	$k_h = 4.0, m_h = 3.4$	0.26	6.47	9.68	15.12	18.01
100	2	$k_h = 4.0, m_h = 3.4$	0.26	8.02	14.93	27.95	31.22
150	2	$k_h = 4.0, m_h = 3.4$	0.26	9.87	20.93	41.44	44.35
200	2	$k_h = 4.0, m_h = 3.4$	0.26	11.56	26.98	52.85	55.68
300	2	$k_h = 4.0, m_h = 3.4$	0.26	14.77	38.62	72.55	74.50
400	2	$k_h = 4.0, m_h = 3.4$	0.26	18.50	48.62	84.90	86.06
500	2	$k_h = 4.0, m_h = 3.4$	0.26	22.02	58.51	92.46	93.10
50	3	$k_h = 2.5, m_h = 3.6$	0.39	6.28	12.90	18.73	24.58
100	3	$k_h = 2.5, m_h = 3.6$	0.39	7.42	23.42	37.29	43.65
150	3	$k_h = 2.5, m_h = 3.6$	0.39	8.97	34.56	56.16	61.55
200	3	$k_h = 2.5, m_h = 3.6$	0.39	10.05	44.68	70.41	74.56
300	3	$k_h = 2.5, m_h = 3.6$	0.39	12.63	61.43	88.36	90.35
400	3	$k_h = 2.5, m_h = 3.6$	0.39	15.41	74.24	95.70	96.42
500	3	$k_h = 2.5, m_h = 3.6$	0.39	17.37	83.40	98.56	98.84

Table 6: Scenario S6 (rare outlier contamination): estimated rejection percentages by sample size and effect level

n_{arm}	h	Effect	Expected $\Delta^{(0)}$	Wilcoxon (%)	Welch t -test (%)	LRT (%)	SPLRT (%)
50	0	$\delta_h = 0.00$	0.00	4.96	4.65	4.68	12.55
100	0	$\delta_h = 0.00$	0.00	4.82	4.88	5.02	6.58
150	0	$\delta_h = 0.00$	0.00	4.96	4.88	4.91	5.42
200	0	$\delta_h = 0.00$	0.00	5.08	4.89	5.01	5.27
300	0	$\delta_h = 0.00$	0.00	4.81	4.99	4.94	5.12
400	0	$\delta_h = 0.00$	0.00	5.00	4.77	4.88	4.98
500	0	$\delta_h = 0.00$	0.00	5.14	5.09	5.09	5.20
50	1	$\delta_h = 0.15$	0.12	22.17	6.10	7.68	15.14
100	1	$\delta_h = 0.15$	0.12	39.28	7.18	8.56	9.24
150	1	$\delta_h = 0.15$	0.12	54.42	8.15	9.98	9.90
200	1	$\delta_h = 0.15$	0.12	66.69	9.82	11.68	11.40
300	1	$\delta_h = 0.15$	0.12	83.44	11.46	14.13	13.68
400	1	$\delta_h = 0.15$	0.12	92.36	14.04	17.96	17.37
500	1	$\delta_h = 0.15$	0.12	96.88	16.62	21.98	21.39
50	2	$\delta_h = 0.25$	0.20	49.92	8.55	13.35	17.63
100	2	$\delta_h = 0.25$	0.20	78.80	11.43	15.25	14.25
150	2	$\delta_h = 0.25$	0.20	92.20	14.73	19.55	17.95
200	2	$\delta_h = 0.25$	0.20	97.50	17.81	24.27	22.52
300	2	$\delta_h = 0.25$	0.20	99.80	23.56	33.51	31.62
400	2	$\delta_h = 0.25$	0.20	99.96	30.24	43.54	41.67
500	2	$\delta_h = 0.25$	0.20	100.00	36.08	51.66	49.88
50	3	$\delta_h = 0.35$	0.28	74.76	11.10	18.32	18.73
100	3	$\delta_h = 0.35$	0.28	96.21	17.47	25.78	22.36
150	3	$\delta_h = 0.35$	0.28	99.52	23.70	34.07	30.58
200	3	$\delta_h = 0.35$	0.28	99.96	29.42	42.66	39.19
300	3	$\delta_h = 0.35$	0.28	100.00	40.94	59.25	56.05
400	3	$\delta_h = 0.35$	0.28	100.00	51.72	71.84	69.16
500	3	$\delta_h = 0.35$	0.28	100.00	60.11	81.16	79.27

Table 7: Scenarios S1–S6: scenario-matched null calibration at effect level $h = 0$

Scenario	n_{arm}	Wilcoxon (%)	Welch t -test (%)	LRT (%)	SPLRT (%)
S1	50	4.94	5.03	5.32	5.37
S1	100	4.72	4.73	5.19	5.21
S1	150	4.98	5.02	5.10	5.11
S1	200	5.16	5.09	5.28	5.30
S1	300	5.00	5.00	4.94	4.94
S1	400	4.95	4.87	4.95	4.94
S1	500	4.68	4.71	4.78	4.78
S2	50	4.83	4.86	5.48	5.64
S2	100	5.28	5.24	5.38	5.49
S2	150	4.95	5.02	5.18	5.24
S2	200	5.02	5.01	4.96	5.00
S2	300	4.69	4.72	4.96	4.95
S2	400	4.96	4.97	4.83	4.84
S2	500	5.02	5.01	4.94	4.95
S3	50	5.08	5.15	5.60	5.77
S3	100	4.75	4.94	5.16	5.19
S3	150	5.13	5.06	5.21	5.26
S3	200	4.88	4.83	4.98	5.00
S3	300	4.85	4.86	5.00	5.01
S3	400	5.04	4.93	5.05	5.04
S3	500	5.10	5.12	5.14	5.14
S4	50	5.00	5.05	5.72	5.97
S4	100	5.05	5.08	5.13	5.18
S4	150	4.94	5.02	5.12	5.18
S4	200	4.99	4.90	5.02	5.04
S4	300	4.96	4.96	5.13	5.11
S4	400	5.24	5.19	5.46	5.47
S4	500	5.02	5.19	5.05	5.01
S5	50	5.06	5.18	5.58	5.80
S5	100	4.86	4.93	5.04	5.15
S5	150	4.91	4.96	5.16	5.19
S5	200	4.90	4.91	5.23	5.28
S5	300	5.04	4.98	5.10	5.11
S5	400	4.85	4.76	4.87	4.88
S5	500	4.98	4.80	5.03	5.02
S6	50	4.96	4.65	4.68	12.55
S6	100	4.82	4.88	5.02	6.58
S6	150	4.96	4.88	4.91	5.42
S6	200	5.08	4.89	5.01	5.27
S6	300	4.81	4.99	4.94	5.12
S6	400	5.00	4.77	4.88	4.98
S6	500	5.14	5.09	5.09	5.20

C Supplementary coded scale interval computation

The likelihood-based coded scale intervals are computed by optimizing the deterministic functional $D_{e_0}(\zeta)$ over sublevel sets of the full criterion $\mathbb{W}_n(\zeta)$ from Section 2. Throughout this appendix, e_0 is the fixed numerical code for the abstract atom $\mathcal{E}$.

For computation it is convenient to use the equivalent arm coordinates

$$(\mu_0, \mu_1, \ell_0, \ell_1), \quad \pi_r = \text{logit}^{-1}(\ell_r).$$

In these coordinates,

$$D_{e_0} = \pi_1(\mu_1 - e_0) - \pi_0(\mu_0 - e_0).$$

The parametric criterion fixes all four coordinates and profiles only the common variance. The empirical likelihood criterion is the sum of the two one-sample empirical likelihood mean deviances

at μ_0 and μ_1 and the two arm-specific Bernoulli deviances at ℓ_0 and ℓ_1 . Thus neither criterion profiles a baseline mean or baseline probability before the coded scale constraint is imposed.

Both reported intervals are ranges of sublevel sets of the same full criterion. The projected interval uses $c = \chi_{4,1-\alpha}^2$, whereas the profile interval uses $c = \chi_{1,1-\alpha}^2$ by the exact image identity in Section 2. Thus their endpoints solve

$$\inf_{\zeta} \{D_{e_0}(\zeta) : \mathbb{W}_n(\zeta) \leq c\} \quad \text{and} \quad \sup_{\zeta} \{D_{e_0}(\zeta) : \mathbb{W}_n(\zeta) \leq c\}.$$

Gaussian working likelihood. Let m_r and $\bar{Y}_r$ be the non-atom count and mean in arm r , let $M = m_0 + m_1$, and let $\text{SSE}_{\text{unres}}$ be the pooled residual sum of squares at the unrestricted fit. For fixed arm logits, write W_A for their Bernoulli deviance. If $W_A \leq c$, the remaining mean criterion satisfies

$$m_0(\mu_0 - \bar{Y}_0)^2 + m_1(\mu_1 - \bar{Y}_1)^2 \leq Q, \quad Q = \text{SSE}_{\text{unres}} \left[\exp\left(\frac{c - W_A}{M}\right) - 1 \right].$$

For the probabilities corresponding to those logits, define

$$B = \pi_1(\bar{Y}_1 - e_0) - \pi_0(\bar{Y}_0 - e_0), \quad V_\pi = \left(\frac{\pi_0^2}{m_0} + \frac{\pi_1^2}{m_1} \right)^{1/2}.$$

The exact conditional extrema over the mean ellipse are $B - \sqrt{Q}V_\pi$ and $B + \sqrt{Q}V_\pi$. Their attaining mean displacements are proportional to $(\pi_0/m_0, -\pi_1/m_1)$ and its negative. The remaining search is therefore only over the compact two-dimensional set of logits satisfying $W_A \leq c$.

For each arm, finite logit bounds are obtained by solving the arm Bernoulli deviance equation at c . The joint Bernoulli sublevel is strictly convex about the unrestricted logits. Along every polar direction from that center there is consequently one boundary radius. We evaluate the bounded angle and radius domain from multiple starting values, refine the leading candidates in both coordinates, and reconstruct the corresponding means from the formula above. The full four-parameter deviance is checked against c within numerical tolerance at each reported endpoint. The analytic reduction over the means is exact, while the remaining two-dimensional optimization is numerical.

Empirical likelihood. In this case there is a simpler exact reduction. Consider arm r with d_r atom observations and non-atom values $Y_{r1}, \dots, Y_{rm_r}$. Let ρ_r and π_r be the atom and non-atom masses, and let q_{rj} be the conditional empirical weights. Symmetry gives individual atom weights ρ_r/d_r , while the non-atom observations have weights $\pi_r q_{rj}$. The ordinary empirical likelihood ratio for the piecewise coded observations in this arm is therefore

$$\left(\frac{n_r \rho_r}{d_r} \right)^{d_r} (n_r \pi_r)^{m_r} \prod_{j=1}^{m_r} q_{rj}, \quad n_r = d_r + m_r.$$

This is exactly the Bernoulli likelihood ratio multiplied by the conditional non-atom empirical likelihood ratio. Moreover, its weighted coded mean is

$$e_0 \rho_r + \pi_r \sum_{j=1}^{m_r} q_{rj} Y_{rj}.$$

After profiling subject to a coded mean difference, the factored full criterion is therefore exactly the ordinary two-sample empirical likelihood criterion for the coded observations in finite samples. Constructing the coded observations piecewise gives one root on each monotone branch of that criterion at cutoff c . The cutoff has one degree of freedom for the profile interval and four degrees of freedom for the projected interval.

These reductions provide numerical evaluations of the optimization problems in Equations (5) and (6) without arbitrary search limits. Under the finite-sample conditions stated in Section 2, their endpoints are finite. An endpoint is reported as unavailable if the numerical calculation fails.

D Supplementary Bayesian posterior predictive checks

This appendix records the posterior predictive checks used for the unadjusted Bayesian two-part model in Section 2.4 (Gelman et al., 1996). A posterior predictive check compares a discrepancy computed from the observed data with the same discrepancy computed from data replicated under posterior draws of the fitted model. Values near zero indicate that the observed discrepancy is unusually large relative to replicated data, while values near one indicate that it is unusually small. These summaries are not calibrated classical p -values.

Let $\vartheta^{(s)}$, $s = 1, \dots, S$, denote posterior draws from the selected Bayesian kernel specification. For arm $r \in \{0, 1\}$, write

$$A_i \mid R_i = r, \vartheta^{(s)} \sim \text{Bernoulli}(\pi_r^{(s)}), \quad \rho_r^{(s)} = 1 - \pi_r^{(s)}, \quad Y_i \mid A_i = 1, R_i = r, \vartheta^{(s)} \sim F_r^c(\cdot \mid \vartheta^{(s)}).$$

Let

$$A = (A_i)_{i=1}^n, \quad n_r = \sum_{i=1}^n \mathbb{1}\{R_i = r\}, \quad \mathcal{I}_r = \{i : R_i = r, A_i = 1\}, \quad m_r = |\mathcal{I}_r|.$$

The atom check replicates $A_i^{\text{rep},(s)}$ for all subjects with $R_i = r$ from the Bernoulli law above. Its discrepancy is

$$T_A(a, \vartheta^{(s)}) = \max_{r \in \{0, 1\}} \left| n_r^{-1} \sum_{i: R_i = r} (1 - a_i) - \rho_r^{(s)} \right|.$$

Thus

$$T_A^{\text{obs},(s)} = T_A(A, \vartheta^{(s)}), \quad T_A^{\text{rep},(s)} = T_A(A^{\text{rep},(s)}, \vartheta^{(s)}).$$

The non-atom check is conditional on the observed non-atom sample sizes. For each arm r and posterior draw s , generate

$$Y_{r,1}^{\text{rep},(s)}, \dots, Y_{r,m_r}^{\text{rep},(s)} \stackrel{\text{iid}}{\sim} F_r^c(\cdot \mid \vartheta^{(s)}).$$

Let $Y_r = (Y_i)_{i \in \mathcal{I}_r}$, let $Y_r^{\text{rep},(s)}$ denote the corresponding replicated sample, and let $\hat{F}_r^c(\cdot; y_r)$ be the empirical CDF of the arm r sample supplied as y_r . For $y = (y_0, y_1)$, define

$$T_C(y, \vartheta^{(s)}) = \max_{r \in \{0, 1\}} \sup_{t \in \mathcal{Y}} \left| \hat{F}_r^c(t; y_r) - F_r^c(t \mid \vartheta^{(s)}) \right|.$$

The observed and replicated discrepancies are

$$T_C^{\text{obs},(s)} = T_C((Y_0, Y_1), \vartheta^{(s)}), \quad T_C^{\text{rep},(s)} = T_C((Y_0^{\text{rep},(s)}, Y_1^{\text{rep},(s)}), \vartheta^{(s)}).$$

For bounded reported score fits, the supremum over $\mathcal{Y}$ is replaced by a maximum over the reported score set $\mathcal{S}$, using the fitted discrete score CDF. For positive support models, the same CDF discrepancy may be evaluated on the log outcome scale with the fitted CDF transformed accordingly. Other discrepancies could target arm-specific means, tail probabilities, or quantiles.

For $k \in \{A, C\}$, the posterior predictive p -value is

$$p_{\text{ppc},k} = \mathbb{P}\left(T_k^{\text{rep}}(\vartheta) \geq T_k^{\text{obs}}(\vartheta) \mid \mathcal{D}_n\right),$$

estimated by

$$S^{-1} \sum_{s=1}^S \mathbb{1}\left\{T_k^{\text{rep},(s)} \geq T_k^{\text{obs},(s)}\right\}.$$

These are Bayesian model checking summaries, not classical significance tests or evidence about between-arm differences. Inference on between-arm differences is based on posterior summaries of the prespecified functionals of the two-part law.

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